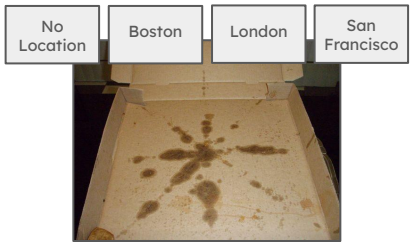
| | 1. Matching with multiple types of disposal bins | 2. Location-specific guidelines | 3. Contamination or damage in the object | 4. Multi-material objects |
|---|---|---|---|---|
| Task | - Analyze the available bins and the size of the openings<br>- Analyze the object and its material, size, and cleanliness<br>- Place the object in the correct bin | - City- or country-specific guidelines vary, e.g., how to recycle glass and soiled items<br>- Identify not only the object but its conditions<br>- Use the provided guidelines to predict object recyclability | - Analyze transformations (cleanliness and structure)<br>- Predict recyclability based on location-specific guidelines | - Analyze the materials present and their method of adhesion<br>- Predict object recyclability based on materials and cleanliness |
| Common Problems | - Often fails to recognize all the openings<br>- Attempts to put large items (e.g., box) into small bins. | - Guidelines often insufficient to determine truth labels | - Difficult to find or generate images that show the same object's transformation of contamination/damage<br>- Guidelines often insufficient to determine truth labels | - Some objects must be separated before recycling (e.g., a glass jar with a metal cap); if not, it is not recyclable. Separability is difficult to assess. |
| Example |   <br>Expected: "none"<br>Result: "left"<br>GPT-4o |  <br>**Without Location:**<br>Expected: yes<br>Result: no<br>**With Boston Guidelines:**<br>Expected: yes<br>Result: yes<br>GPT-4o-mini | **Before:** **After:**<br>  <br>Expected: yes<br>Result: yes<br>Expected: yes<br>Result: no<br>Claude 3.5 |  <br>Expected: yes<br>Result: yes<br>GPT-4o |